

Advanced Visuals in Power BI

Assignment Questions — Complete Solutions

Dataset: Sales_dataset

How to use this document: Questions 2, 3, 5, 7, 8, 9, 11, 12, 14 and 15 are hands-on Power BI tasks. Each includes exact click-by-click steps and any DAX formulas needed so you can reproduce the result in Power BI Desktop with the Sales_dataset in a few minutes. Conceptual questions (1, 4, 6, 10, 13) include definitions, comparisons and real-world examples suitable for direct submission.

Q1. Difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI

All three narrow down the data shown in visuals, but they differ in scope, complexity and how the end user interacts with them.

Aspect	Basic Filter	Advanced Filter	Slicer
Location	Filters pane (Visual/Page/Report)	Filters pane (Advanced mode)	On-canvas visual
Complexity	Simple checklist / search-select values	Conditional logic (AND/OR, contains, greater than, etc.)	Simple, click-to-select values
Visibility to end user	Hidden in the Filters pane unless pinned	Hidden in the Filters pane	Always visible on the report canvas
Best for	Quickly including/excluding known values	Rule-based filtering, e.g. 'Sales > 50000 AND Region ≠ West'	Interactive, visual, end-user-friendly filtering
Example	Show only Year = 2023	Show Sales greater than 10000 but less than 50000	A Region button list users click to filter the page

Q2. Apply a Basic Filter to display only 2023 sales

1. Load the Sales_dataset into Power BI Desktop (Home → Get Data → Excel/CSV) and load it into the data model.
2. Select any visual on the report canvas (or click on empty canvas for a report-level filter).
3. Open the Filters pane (View → Filters, or the pane on the right).
4. Drag the Order Date (or Year) field into the Filters pane.
5. Power BI shows it as a Basic Filter by default — tick the checkbox for 2023 only (or use the search box to find '2023').
6. Click Apply filter.

Note: If your date field isn't already broken into Year/Month/Day, right-click the date field in the Fields pane and choose 'New column' with YEAR([Order Date]), or simply expand the built-in date hierarchy and filter on the Year level.

Q3. Pie Chart of Sales by Product + Region Slicer

7. Insert a Pie Chart: Visualizations pane → Pie chart icon.
8. Drag Product into the Legend field well.
9. Drag Sales (or your Sales Amount measure) into the Values field well — it will default to Sum of Sales.

10. Format the pie chart: turn on Data labels (Format pane → Data labels → On) to show percentage/value per slice.
11. Insert a Slicer: Visualizations pane → Slicer icon, and place it beside the pie chart.
12. Drag Region into the slicer's Field well.
13. In the Format pane for the slicer, set Slicer settings → Style to 'Tile' or 'Dropdown' for a cleaner look.
14. Click through the Region slicer values — the pie chart's Sales-by-Product breakdown should update automatically because slicers filter all visuals on the page by default.

Q4. Visual-Level Filter vs Page-Level and Report-Level Filters

Filter Type	Scope	Where applied
Visual-Level Filter	Only the single visual it is applied to	Filters pane → 'Filters on this visual'
Page-Level Filter	Every visual on the current report page	Filters pane → 'Filters on this page'
Report-Level Filter	Every visual on every page of the report	Filters pane → 'Filters on all pages'

In short, the three filter types form a hierarchy of scope: a report-level filter is the broadest (whole report), a page-level filter narrows it to one page, and a visual-level filter is the most granular, affecting only one chart or table — useful when, say, you want a single KPI card to always show total sales regardless of other filters on the page.

Q5. Report-Level Filter — show sales only from India

15. Click on any empty area of the report canvas so no single visual is selected.
16. Open the Filters pane.
17. Drag the Country field into the 'Filters on all pages' section.
18. Set the filter type to Basic filtering and check only India.
19. Click Apply filter.

Every visual on every page — pie charts, tables, cards, maps — will now show figures for India only, since a report-level filter overrides page- and visual-level scope by applying before them.

Q6. Role of Drill-Down in Power BI

Drill-down lets a user move from a summarized (high) level of a hierarchy to progressively more detailed levels within the same visual — for example, from Country totals down to Region, then down to individual Product Category — without building a separate chart for each level.

It improves data exploration because:

- One visual serves multiple levels of detail, reducing report clutter.
- Users can self-serve: they explore top-down, at their own pace, instead of the report author anticipating every breakdown in advance.
- It preserves context — the chart's type, formatting and filters stay the same as you go deeper, so comparisons remain visually consistent.
- It supports 'expand all' as well as 'drill into' one branch, giving both a big-picture and a focused view from the same object.

Q7. Hierarchy Drill-Down: Country → Region → Product Category

20. Choose a visual that supports hierarchies well, e.g. a Clustered Column Chart or a Matrix.
21. Drag Country into the Axis (or Rows) field well.

22. Drag Region into the same Axis/Rows field well, directly below Country — Power BI automatically creates a hierarchy (Country → Region).
23. Drag Product Category into the same field well, below Region, extending the hierarchy to Country → Region → Product Category.
24. Drag Sales into the Values field well.
25. On the visual header, click the down-arrow drill icon (or right-click a column/bar → Drill Down) to move one level deeper, and the up-arrow to go back.
26. Use the double-down 'Expand' icon to show all Region totals (or all Product Category totals) at once, broken out under each Country, instead of drilling into one branch at a time.

Q8. Drill-Through: Product → detail page with region-wise sales

27. Add a new report page and rename it, e.g. 'Product Details'.
28. On this new page, open the Visualizations pane → Filters section and drag the Product field into the well marked Drill-through, at the top of the Filters pane.
29. Power BI automatically adds a Back button to the page — keep it, so users can return to the source page.
30. Build the page's content: add a table or bar chart with Region on the axis/rows and Sales in values, plus any KPI cards you want (Total Sales, Total Quantity, etc.) for that product.
31. Go back to the source page (e.g. main dashboard) that has a visual containing Product (a bar chart, table, etc.).
32. Right-click a specific product's data point in that visual → Drill through → Product Details.

Power BI navigates to the 'Product Details' page, automatically filtered to the product you clicked, and shows the region-wise sales breakdown for just that product.

Q9. Slicer for Product Category interacting with the Drill-Down hierarchy

33. Add a Slicer visual (Visualizations pane → Slicer icon) to the same page as the Country → Region → Product Category hierarchy chart.
34. Drag Product Category into the slicer's Field well.
35. Select one or more categories in the slicer, e.g. 'Electronics'.

Because slicers and drill-down hierarchies both operate on the same underlying data model, the interaction works as follows:

- The chart's data is first filtered to the selected Product Category by the slicer.
- The drill-down hierarchy (Country → Region → Product Category) then only shows Countries/Regions that have sales in that category, and totals reflect the filtered category only.
- If you drill all the way down to the Product Category level, the chart will effectively show just the category (or categories) chosen in the slicer, since deeper levels and the slicer selection compound (both are AND conditions).

Note: If you want the slicer to affect other pages too, right-click the slicer → 'Sync slicers' and enable it across the relevant pages.

Q10. Drill-Down vs Drill-Through — with a real-world example

Aspect	Drill-Down	Drill-Through
What it does	Moves through levels of ONE hierarchy inside the SAME visual	Jumps to a completely different, often more detailed, report page
Data shown	Same measure, increasing granularity (e.g. Country → Region → City)	A different set of visuals/measures, filtered to the selected item
Navigation	Stays on the same page	Navigates to another page (with a Back button)

Aspect	Drill-Down	Drill-Through
Typical use	Exploring a hierarchy quickly, e.g. Year → Quarter → Month	Investigating one specific entity in depth, e.g. one Customer's full order history

Real-world example: a Sales Manager looking at a national dashboard could drill down from Country → Region → City on a map or bar chart to see where in the country sales are concentrated (same visual, more detail). If she then wants a deep dive on one underperforming City — its top products, its monthly trend, its top customers — she would right-click that city and drill through to a dedicated 'City Deep-Dive' page built specifically for that level of analysis.

Q11. Drill-Down: Year → Quarter → Month → Day

36. Use a Line Chart or Column Chart to show sales trends over time.
37. Drag Order Date into the Axis field well. If Power BI's automatic date hierarchy is enabled, it already creates Year → Quarter → Month → Day for you.
38. Drag Sales into the Values field well.
39. Click the drill-down arrow (▼) on the visual header to move from Year down to Quarter, then Month, then Day, one level at a time.
40. Alternatively, click the 'Expand' (⇅) icon to show all Quarters (or all Months) for every Year side-by-side instead of drilling into a single year.
41. Use the up-arrow (▲) to go back up a level at any time.

Note: If the automatic date hierarchy is turned off (File → Options → Data Load → uncheck 'Auto date/time'), build a manual hierarchy instead: create calculated columns for Year, Quarter, Month and Day from Order Date, then drag them into the Axis well in that order to form a custom hierarchy.

Q12. Measure-Based Filter — products with total sales exceeding ₹50,000

Step 1 — Create the measure:

```
Total Sales = SUM('Sales_dataset'[Sales Amount])
```

Step 2 — Apply the filter using the measure, not the raw column:

42. Select the visual (e.g. a table or bar chart of Product vs Total Sales).
43. Open the Filters pane and drag the Total Sales measure (not the Sales Amount column) into 'Filters on this visual'.
44. Set the filter type to Advanced filtering → 'is greater than' → 50000.
45. Click Apply filter.

This filters on the aggregated total per product, so a product with many small transactions that together exceed ₹50,000 is correctly included — filtering on the raw Sales Amount column instead would filter individual transaction rows rather than product totals, giving the wrong result.

Q13. 'Keep All Filters' option in Drill-Through

When you right-click to drill through, a small dialog can offer a 'Keep all filters' toggle (also configurable per drill-through page). It controls whether filters/slicers active on the source page travel with you to the destination page, in addition to the drill-through field itself.

- Keep all filters ON: the destination page is filtered by the drill-through field AND by every other filter/slicer that was active on the source page (e.g. a Year slicer selection carries over).
- Keep all filters OFF (default): only the drill-through field's value filters the destination page; any other filters/slicers on the source page are ignored.

This matters for user interaction because it determines whether the detail page reflects the exact context the user was viewing (consistent, but can over-filter and hide relevant rows) or always resets to a broader, drill-through-field-only view (simpler, but loses context like a selected date range).

Q14. Custom Drill-Through page: Region → sales trends over time

46. Create a new page named 'Region Trend Detail'.
47. In the Filters pane on this page, drag Region into the Drill-through well at the top.
48. Keep the auto-generated Back button.
49. Add a Line Chart: Order Date (using the Year → Quarter → Month hierarchy) on the Axis, Sales on Values — this becomes the time-trend visual for whichever region the user drilled from.
50. Optionally add supporting visuals: a card showing Total Sales for the region, and a table of Top Products in that region, for richer context.
51. On the source page, right-click a Region value in any chart/table → Drill through → Region Trend Detail.

The result: selecting any region takes the user straight to a page showing that region's sales trend over time, with a one-click Back button to return to the overview.

Q15. Design a fully interactive dashboard (Slicers + Filters + Drill-Down + Drill-Through)

Recommended page layout for a single, cohesive Sales dashboard:

52. Page 1 — Overview: KPI cards (Total Sales, Total Orders, Avg Order Value) at the top; a Clustered Column Chart with the Country → Region → Product Category drill-down hierarchy (from Q7); a Pie Chart of Sales by Product (from Q3); a Line Chart with the Year → Quarter → Month → Day drill-down (from Q11).
53. Slicers panel on the left: Region slicer, Product Category slicer, and a Date range slicer — all synced (Format → Edit interactions, or Slicer → Sync slicers) so a change in one instantly updates every visual on the page.
54. Report-level filter: Country = India applied in 'Filters on all pages' (from Q5), or removed/parametrized if the dashboard should cover all countries — adjust to the actual scope required.
55. Visual-level filter example: on the KPI card showing 'Top Category Sales', add a measure-based filter (Total Sales > 50000, from Q12) so it only ever highlights genuinely significant categories.
56. Drill-through pages: 'Product Details' (from Q8) reachable from the Product pie chart / bar chart, and 'Region Trend Detail' (from Q14) reachable from any Region visual — both with Back buttons and 'Keep all filters' set deliberately (from Q13) based on whether cross-page context should persist.
57. Add bookmarks or a page navigator (buttons) at the top for quick jumps between Overview, Product Details and Region Trend Detail, so the dashboard feels like one connected app rather than separate pages.
58. Finish with consistent formatting: a shared color theme (View → Themes), aligned card sizes, and data labels turned on for key charts, so the dashboard is both interactive and visually polished.

This combination — synced slicers for instant filtering, layered basic/report/visual filters for scope control, drill-down hierarchies for progressive detail within a chart, and drill-through pages for deep dives — gives users a single dashboard that supports both quick, high-level scanning and detailed, self-directed exploration.